LESSON PLAN

WORKSHOP PRACTICES (PRACTICAL) (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Guided missile control system, introduction functions and types of control systems (Aerodynamic controls and thrust controls) and classifications
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Guided missile control system, introduction functions and types of control systems (Aerodynamic controls and thrust controls) and classifications of aerodynamic and thrust control.
5. Specific Objectives	23.1.1.1 Trainee will be able the basic knowledge of air craft armaments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of control surface diagrams and thrust vectoring mechanisms; Safety: Handle diagrams carefully; Instructor supervision: Monitor understanding of control classifications.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a guided missile control system?
(b) Name two main types of control systems used in missiles.
(c) How does a control system differ from a guidance system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Guided Missile Control Systems • Basic Functions of Control Systems • Importance of Stability and Maneuverability	L & Ds	PPT, IB	15 min
(b)	• Aerodynamic Control Systems Overview • Classification of Aerodynamic Controls • Functioning of Control Surfaces	L & Ds	PPT, PDF	20 min
(c)	• Thrust Control Systems Overview • Classification of Thrust Controls • Comparison of Control Types	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between aerodynamic and thrust control systems.
(b) Classify the types of aerodynamic controls used in missiles.
(c) Describe how thrust vectoring contributes to missile guidance.

12. Home Assignment and Follow Up	Prepare a brief note on the types of guided missiles used in the Navy for the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of guided missiles used in Navy.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min